

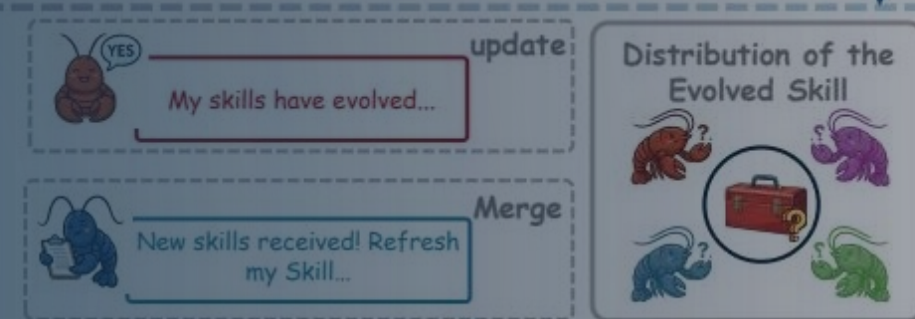
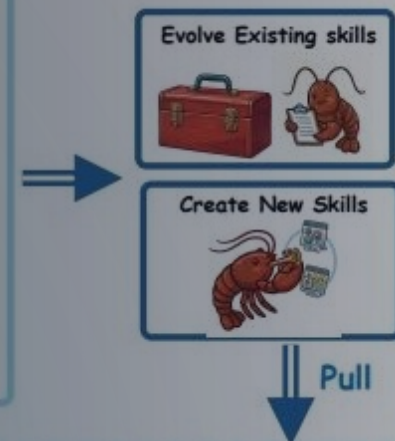
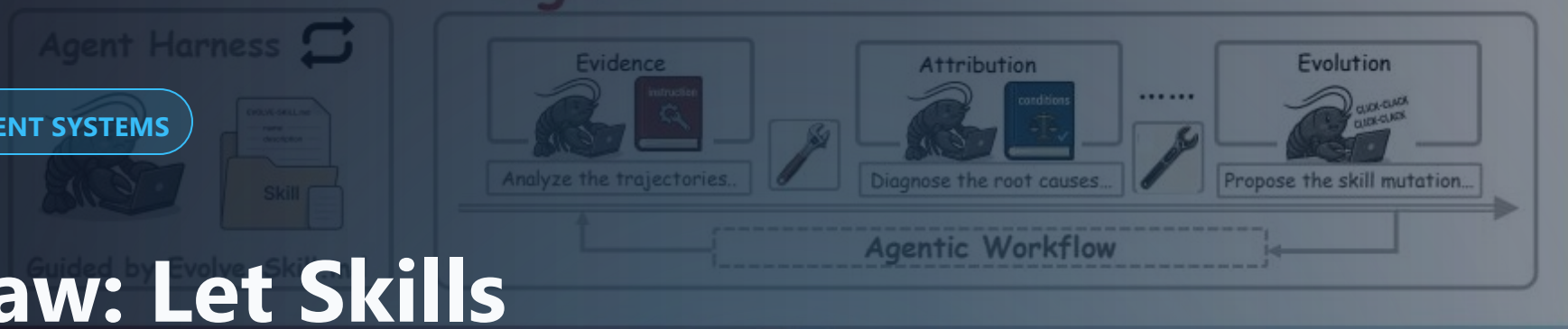
## MULTI-USER AGENT SYSTEMS

# SkillClaw: Let Skills Evolve Collectively with Agentic Evolver

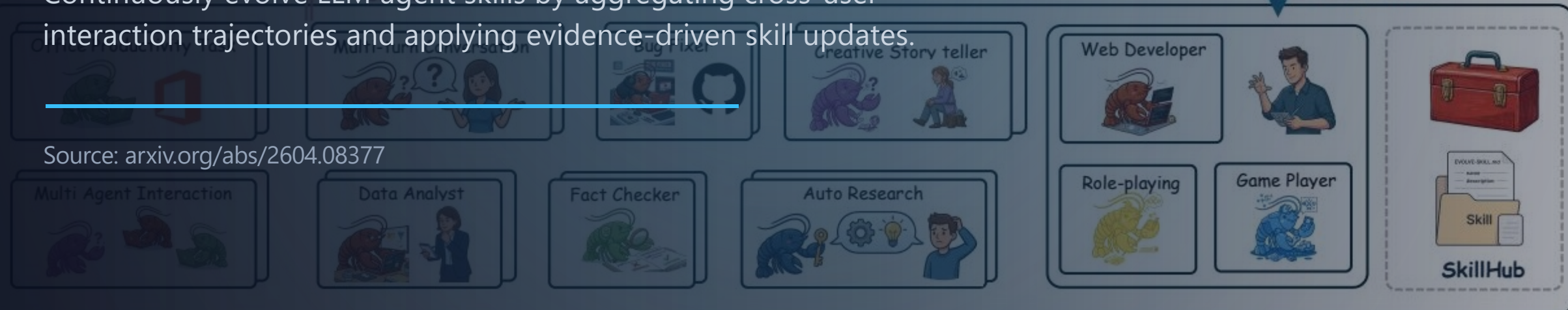
Continuously evolve LLM agent skills by aggregating cross-user interaction trajectories and applying evidence-driven skill updates.

Source: [arxiv.org/abs/2604.08377](https://arxiv.org/abs/2604.08377)

## Agentic Evolver



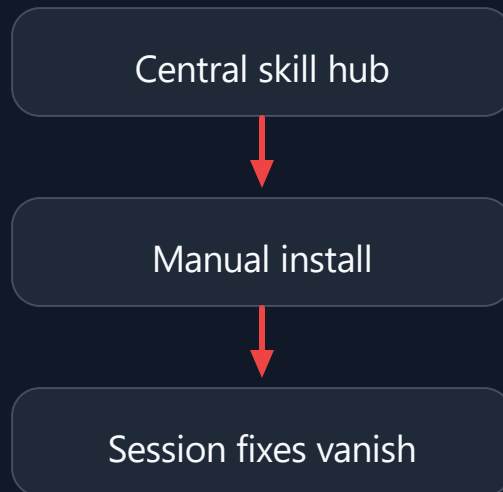
Verification



# Agent Skills Remain Static After Deployment

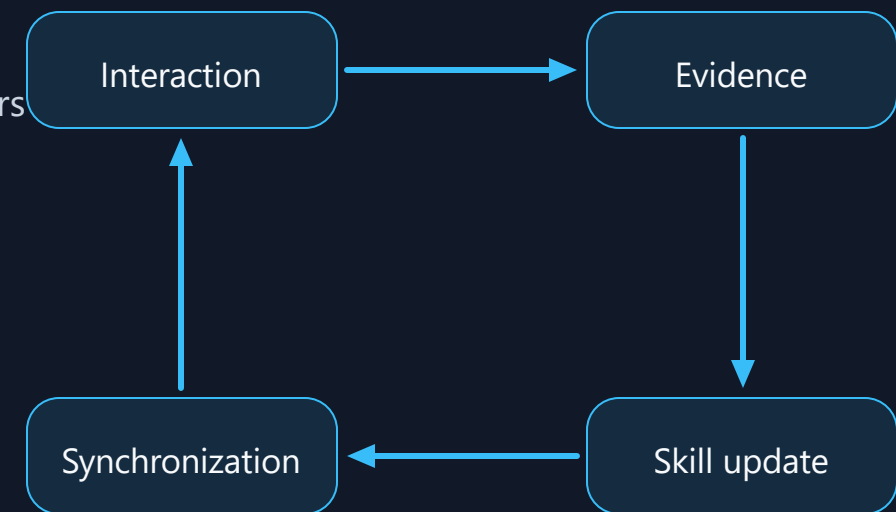
Users rediscover solutions independently because runtime experience does not update the shared skill library

## Static skill paradigm



- Skills remain static after deployment
- Failures and recoveries recur across users
- Memory methods stay tied to instances

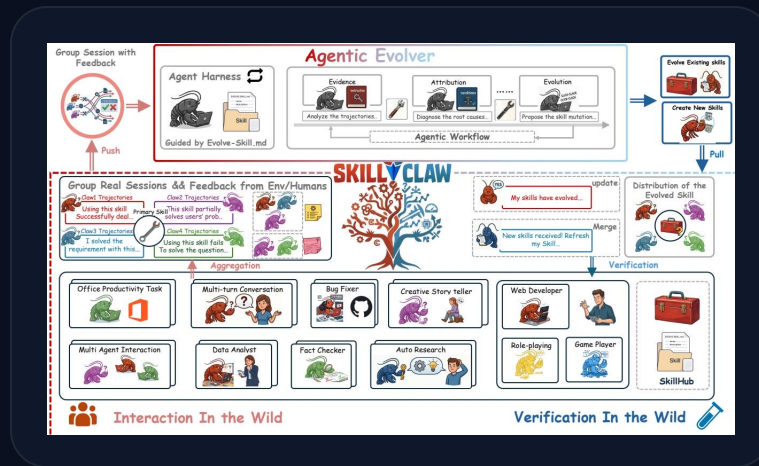
## Evolving skill paradigm



Runtime interaction contributes to a shared, continuously improving ecosystem.

# Closed-Loop Pipeline: From Multi-User Interaction to Shared Skill Evolution

Four stages connect many users to a synchronized, improving skill repository



Architecture figure reference

**Multi-user Interaction**  
full action–feedback causal chains

**Evidence**  
Aggregate structured session trajectories

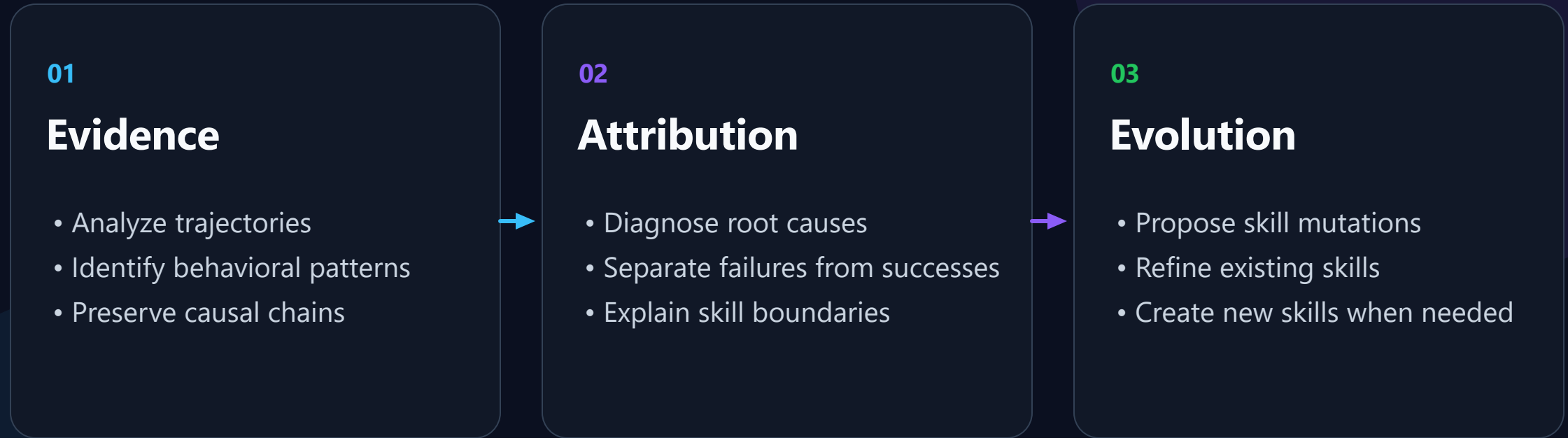
**Attribution**  
Diagnose root causes of failures + successes

**Distribution**  
Verify, merge, and synchronize to agents

**Evolution**  
Propose skill mutations or new skills

# Agentic Evolver: Evidence → Attribution → Evolution in Three Stages

The update mechanism is open-ended agent reasoning over interaction evidence — not predefined rules



**Autonomous evolver reasons over evidence before verified updates enter the shared repository**

# Cross-User Trajectory Aggregation Exposes Skill Behavioral Boundaries

Different Claw-style systems exercise the same skill under diverse contexts, producing complementary signals



**A single user rarely generates enough signal to separate a generalizable improvement from an idiosyncratic fix.**

# Case Study: Multimodal Creative Pipeline Skill

1 / 6 accepted

Six days of evolution attempts illustrate conservative verification that prevents regressions

| Day | Date  | Outcome  | Rationale                                                                                                   |
|-----|-------|----------|-------------------------------------------------------------------------------------------------------------|
| 1   | Day 1 | Accepted | validate-tmp-workspace-inputs; checks /tmp_workspace inputs & environment setup<br>Upgrade to new best pool |
| 2   | Day 2 | Rejected | multimodal-input-validation-and-setup<br>Keep current best pool                                             |
| 3   | Day 3 | Rejected | multimodal-creative-task-pipeline; extract from PDF/video/image<br>Keep current best pool                   |
| 4   | Day 4 | Rejected | Added image classification, visual generation, structured validation<br>Keep current best pool              |
| 5   | Day 5 | Rejected | Creative pipeline + per-file fail-fast validation<br>Keep current best pool                                 |
| 6   | Day 6 | Rejected | Extended PDF-to-poster / document-to-visual paths<br>Not admitted to next cycle                             |

Note: source brief provides day numbers, not calendar dates; the date field preserves the provided day-by-day sequence.

# Case Study: Git Push Auth-Fallback Skill

3 / 6 accepted

Accepted updates appear for the first three days, followed by plateau and rejected no-improvement changes

| Day | Date  | Outcome  | Rationale                                                                                        |
|-----|-------|----------|--------------------------------------------------------------------------------------------------|
| 1   | Day 1 | Accepted | Safe fallback instead of blocking on push failure<br>Add to Safety best pool                     |
| 2   | Day 2 | Accepted | Unified patch/bundle filenames; reduced inconsistency<br>Keep updated best pool                  |
| 3   | Day 3 | Accepted | Auth-alternative paths & secrets audit; fixed subshell pitfalls<br>Keep current best pool        |
| 4   | Day 4 | Rejected | Same-pool retest confirmed current pool as best<br>No new update counted toward accepted updates |
| 5   | Day 5 | Rejected | Added “push hang treated as auth failure” ; no improvement<br>Keep current best pool             |
| 6   | Day 6 | Rejected | Added identity config & filename consistency; no improvement<br>Not admitted to next cycle       |

Note: source brief provides day numbers, not calendar dates; the date field preserves the provided day-by-day sequence.



# Key Takeaways: Three Properties That Distinguish SkillClaw

Implications for continuously improving agent systems

01

## Collective evolution

Individual interactions improve a shared skill ecosystem.

02

## Fully automatic

Runtime interaction drives updates without manual curation.

03

## Agentic evolution

Open-ended reasoning produces evidence-driven skill changes.

**For agent builders: preserve interaction evidence, attribute behavior across users, and verify skill mutations before distributing them back to the ecosystem.**